

LinkedIn Carousel PDF Generator (Ollama + HTML-to-PDF + Google Drive + Gmail)

Generate fully branded LinkedIn carousel PDFs automatically using AI, convert them into LinkedIn-ready document posts, upload them to Google Drive, log every generation in Google Sheets, and receive the final PDF by email.

Overview

This n8n workflow transforms a simple topic submission into a complete LinkedIn carousel PDF ready for publishing.

The user fills out a form with:

- Carousel topic
- Brand name
- Target audience
- Tone of voice
- Number of content slides

An Ollama-powered AI model generates structured carousel content in JSON format. The workflow then builds a visually branded HTML presentation, converts it into a square-format PDF, uploads it to Google Drive, logs the generation in Google Sheets, and sends the final document via Gmail.

Perfect for:

- LinkedIn creators
- Personal brands
- Marketing agencies
- Coaches and consultants
- Content teams
- Social media managers

Features

- ✓ AI-generated LinkedIn carousel content
- ✓ Custom topic, audience, and tone
- ✓ Automatic title slide, content slides, and CTA slide
- ✓ Branded HTML carousel design
- ✓ Square 1080×1080 LinkedIn-ready pages
- ✓ PDF generation using Chromium/Puppeteer

- ✓ Google Drive upload
 - ✓ Google Sheets logging
 - ✓ Gmail delivery
 - ✓ Fully self-hosted using Ollama
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Workflow Architecture

1. Form Trigger

Collects:

- Carousel Topic
 - Brand Name
 - Target Audience
 - Tone
 - Number of Content Slides
-

2. AI Content Generation (Ollama)

The AI Agent uses Ollama (Llama 3.1) to generate a structured JSON carousel.

Output includes:

```
[
  {
    "slide_number": 1,
    "type": "title",
    "headline": "Powerful Hook",
    "subtitle": "Supporting statement"
  }
]
```

Each slide contains:

- Headline
 - Subtitle
 - Emoji
 - Bullet points
 - Slide type
-

3. Parse JSON

A Code node:

- Validates AI output
 - Removes markdown wrappers
 - Parses JSON safely
 - Handles malformed responses
-

4. Build HTML Carousel

Generates a complete multi-page HTML document.

Includes:

- Brand colors
- Typography
- Decorative backgrounds
- Slide numbering
- Swipe indicators
- CTA slide

All styling is embedded inline for reliable PDF rendering.

5. Generate PDF

Uses Puppeteer through html-pdf-node to render the carousel as a PDF.

Output:

- Square pages (1080×1080)
 - High-quality PDF
 - Binary file for further processing
-

6. Upload to Google Drive

Automatically uploads the generated PDF to a specified Google Drive folder.

Returns:

- File ID
 - View URL
 - Download URL
-

7. Log to Google Sheets

Stores generation history including:

- Date
- Topic
- Brand
- Slide count
- File name
- Drive link
- Drive file ID
- PDF size

Useful for analytics and reporting.

8. Send Email Notification

Sends an HTML email containing:

- Topic
 - Slide count
 - File name
 - Google Drive link
 - LinkedIn posting instructions
-

Required Services

Ollama

Install and run Ollama locally:

```
curl -fsSL https://ollama.com/install.sh | sh
ollama pull llama3.1
```

Configure the Ollama credential in n8n.

Google Drive OAuth2

Required for:

- Uploading PDFs
 - File storage
-

Google Sheets OAuth2

Required for:

- Generation logging
 - Reporting
-

Gmail OAuth2

Required for:

- Sending completed carousel notifications
-

Server Requirements

Install html-pdf-node

```
npm install html-pdf-node
```

Docker installation:

```
docker exec -it n8n npm install html-pdf-node
```

Install Chromium

The PDF renderer requires Chromium.

Example Alpine Linux installation:

```
apk add chromium
```

Verify the Chromium path matches:

```
/usr/bin/chromium-browser
```

Update the path in the PDF generation node if necessary.

Customization

Inside the **Build HTML Slides** node you can customize:

```
const BRAND_COLOR = '#0A66C2';  
const FONT = 'Arial, sans-serif';
```

Recommended changes:

- Brand color
 - Typography
 - Background design
 - Slide decorations
 - CTA messaging
-

Example Use Cases

Personal Branding

Create educational LinkedIn carousels automatically.

Marketing Agencies

Generate client carousel content at scale.

Coaches & Consultants

Turn expertise into engaging document posts.

Content Teams

Standardize carousel production across brands.

Output

The workflow produces:

- LinkedIn-ready PDF carousel
- Google Drive file
- Google Sheets log entry
- Email notification

Ready for immediate upload as a LinkedIn Document Post.

Notes

- Designed for self-hosted n8n environments.
- Uses local Ollama models for privacy and reduced AI costs.
- Requires Chromium/Puppeteer support for PDF rendering.

- Supports dynamic branding and slide counts.

License

MIT License

Feel free to modify and extend the workflow for your own projects and clients.